WAKISSHA
MARKING GUIDE
Uganda Advanced Certificate of Education
BIOLOGY P530/2

SECTION A (40 MARKS)

1. a) (Check at the back)

b) Differences

Potassium ion concentration	Rate of sugar consumption
- at 0% lowest	- at 0% higher
- From 0% to 10% increases rapidly	- Increases gradually/less rapid
- Between 10% and 30% increases slowly/ almost constant	- Increase is gradual
- Between 30% and 70% decreases gradually	- Increase is slow
- at 30% there's highest Potassium ion concentration	- highest is at 70%

Any 5 each @1mark

Similarities
- both lowest at 0% oxygen concentration
- both between 0% and 10% (oxygen concentration increase is rapid
- both reach a maximum
- both between certain oxygen concentration remains almost constant

Any 5 each @1mark

c) i) This due to the fact that there has to be a certain concentration; of Potassium ions inside the roots for the ionic gradient.

Movement of ions into roots can occur passively; i.e. without the expenditure of energy; passive transport doesn't require oxygen; so at zero concentration movement of ion into the roots occurs passively though there is low movement;

@ 1mark = 4 marks

ii) Increase in oxygen concentration increases the movement of potassium ions; this is due to the fact that movement of ions occurs against a concentration gradient; and this requires oxygen; Oxygen is needed is respiration (metabolism) to release energy;

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Between 10% and 30% oxygen concentration, the potassium ion movement remains constant;due to all the potassium ion gates being open;hence no observed increase in the entry of ions;into the roots because the Oxygen concentration is Optimum;for the opening of all potassium ion gates;Between 30% to 70% the ion concentration decreases; probably due to the ions being transported away;or the movement of the ions has been reduced due to other factors;e.g. increased temperatures

@ ½ mark
= 6 marks

iii) Increase in Oxygen concentration;increases the respiratory rate (metabolic rate);thus sugars have to be respired in presence of Oxygen;(aerobic respiration to release energy;for the movement of Potassium ions thus an increase in the rate of sugar consumption.

@ 1mark
= 4 marks

d)
- temperature
- Enzyme concentration
- presence of inhibitors and poisons

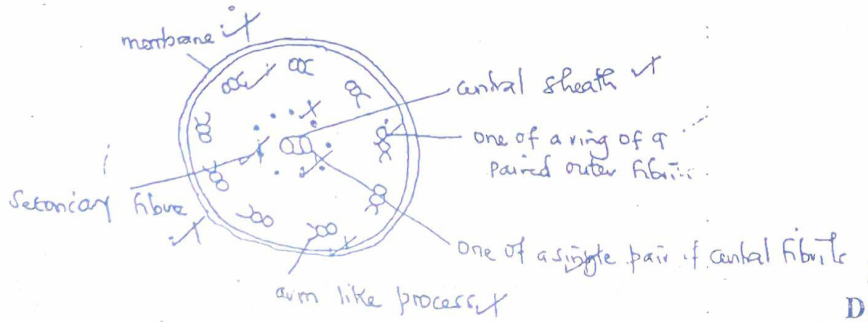
Any 2
@ 1 mark
2 marks

e) The potassium ion concentration would decrease;up to a point and remain constant; high Oxygen concentrations are metabolic poisons;so this will reduce the movement of the ions;The rate of sugar consumption would increase slightly;and then remain constant;because oxidation of sugars releases heat energy;which will result in an increase in the temperatures;within the roots providing unsuitable conditions;for the working of enzymes;and this will consequently reduce the rate of sugar consumption;thus affect the movement of the ions.

max 10,
@1/2mark,
Total 5marks

2. Structure and function of Cilia.

Structure and function of cilia



Drawing = 06marks
Description = 3marks

Cilia are about 5-10μm long and 0.3μm in diameter, this makes it slender enabling it to beat in a rhythmical bringing about movement of the organism or materials towards a certain structure.

It has a 9 microfibrils at the periphery surrounding 2 inner/central micro fibrils hence the 9+2 micro fibril arrangement which supports the cilium.

Description @$^1/_2$mark

b)

Electron Microscope Advantages	Light Microscope Disadvantages
- Greater magnification	- lower magnification
- Greater depth of field	- lesser depth of field
- No complex lighting required	- may involve phase contrast or other complex lighting.
- Higher resolving power	- lower resolving power

Disadvantages	Advantages
- Expensive to buy	- cheap
- Expensive to run	- cheap
- Black and white image	- colored
- Specimen dead	- Specimen living or dead
- Vacuum required	- no vacuum required
- Lengthy, expert preparation required	- Preparation short and simple
- Large requiring Special room	- Small and portable
- Affected by magnetic fields	- unaffected by magnetic fields

Any 7

@ 1mark (should show comparison)

Total = 7marks

c)

Structure	Plant	Animal
Chloroplast	present	absent
Cellulose cell wall	present	absent
Centrioles	Absent	present
Microvilli	Absent	present

Choose any 3, well explained, @ 1mark = 3marks

3. a)

Bond	Description	Example
- hydrogen bond	formed between hydrogen and a more electronegative atom e.g. oxygen and nitrogen.	joining DNA strands
- glycosidic bond	formed between two simple sugars e.g. glucose and fructose in the formation of disaccharides e.g. sucrose	carbohydrates
- Ester linkages	formed between glycerol/(alcohol) and a fatty acid in the formation of a triglyceride .	triglyceride or lipids

- Peptide bond	formed between amino acids in the formation of a peptide	Proteins
- Covalent bond	found in between atoms/elements of any biological molecule	C - H C = O of any monomer of biological molecule
- Ionic bond	these are formed between polar groups of two different amino acids	information of secondary and tertiary structure of proteins.
- disulphide bonds	formed between two sulphur groups of biological molecules	in between 2 amino acids containing sulphur

Bond = ½ mark
Description = ½ mark
example = ½ mark
12 marks

b) In a large population, there is random mating;so in breeding;is always avoided;which results in production of off springs with different genetic composition;from those of parents and themselves due to sexual reproduction;Sexual reproduction always result into off springs with genetic constitution different from parents and themselves due to

- fusion of parental gametes;which are different genotypically
- random distribution of chromosomes;on the metaphase plate during meiosis 1
- exchange of genetic materials;during crossing over
- separation of homologous chromosomes;during meiosis 1.

1@1mark,
Total = 8marks

4. a) Limiting factor

When a process, photosynthesis, is affected;by more than one factor;its rate will be determined;by that factor which is closest;to its minimum concentration;such a factor is the limiting factor;@ ½ **mark,**

3marks

b) Role in Photosynthesis

i) H_2O

Water is needed in the light stage; Photolysis of water; molecules leads to the formation of H+;electrons;and hydroxyl ions;OH^+ eventually will combine to produce a bi product Oxygen; The H^+ will add to $NADP^+$;to

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- Peptide bond	formed between amino acids in the formation of a peptide	Proteins
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Description = ½ mark
example = ½ mark
12 marks

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1@1mark,
Total = 8marks

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form $NADPH_2$; a hydrogen carrier which will go to the dark stage to provide H^+ and energy; The elections produced from water;will pass to stabilize;PSII as the elections fall back to PSII they lose energy; which is used to add ADP to inorganic Phosphate forming ATP;(another store for energy.

@ ½ mark,
max = 5marks

ii) Light

Light is energy which is needed to synthesize chemical energy inform of ATP;and $NADPH_2$; Light energizes electrons;in Pigment systems I and II boasting them to higher energy levels;on falling back to lower energy levels the electrons lose energy; and it's this energy used to add an inorganic;phosphate (pi) to ADP;forming a reservoir of energy called ATP. Light energy is used in the lysis of water; molecules to produce H+; which will be used to synthesize $NADPH_2$;and electrons which will synthesize ATP;.

@ ½ mark,
Max = 4marks

c) Adaptation of shady plans to obtaining light.

- leaves are broad and flat.
- leaves are thin to allow passage of light.
- high concentration of chlorophyll to trap as much light.
- Cuticle and epidermis are transparent.
- Chloroplast are able to orient themselves to trap as much light
- Mosaic pattern of leaves to prevent Obstruction of each other
- Etiolation
- Climbing the tall plants
- Are epiphytes

Any 8
@ 1mark,
max 8marks

5. a) i) Composition of lymph

- dissolved foods
- lymphocytes
- blood plasma
- bacteria, viruses
- cellular wastes like urea, co_2
- dissolved gases e.g. O_2 & CO_2

Composition
Any 3 @ ½ mark

Formation of lymph

Blood reaching the capillary bed;has a lot of pressure;which forces th small molecules;within blood to pass from the capillaries into the cellular space;forming tissue fluid; However continual Oozing of blood constituents due to pressure;forces those in the cellular space;to move back into the venules back to the general circul ation or into the lymptiatic capillaries;or into the lymphatic system;this forms lymph;

Description 5marks,
@ ½ mark

ii) Functions of lymph
- fluid recovery
- immunity
- lipid absorption

@ 1½ mark

b) Hormones controlling blood pressure.

Angiotensin II;is a vasoconstrictor;that raises the blood pressure;drugs such as angiotensin converting enzyme inhibitors lower angiotensin II and blood pressure;

Aldosterone; promotes retention of sodium ions by the kidneys;and hence water retention;thereby promoting high blood volume and pressure;

Atrial natriuretic peptide (ANP);secreted by the heart, antagonizes aldosterone; It increases sodium ion secretion by the kidneys;thus reducing blood pressure and volume; it is a vasodilator,

ADH;promotes water retention;but also is a vaso constrictor;and raises blood pressure;

Adrenalin and noradrenaline;promotes vasoconstriction; and raises blood pressure;

During exercise; they cause vasodilation;increasing blood flow to the heart muscles;and skeletal muscles;

@ ½ mark
Total = 12marks

6. a) Support in plants by;

i) Mechanical strengthening.
- some cells are strengthened with lignin as well as cellulose
- collenchyma cells are heavily lignified and sclerenchyona.-xylem is a dead tissue which later form wood in woody plants.
- In dicotyledonous stems the mechanical tissue forms a cylinder close to the perimeter which provides resistance to horizontal force e.g. wind
- In roots, the central core is strengthened tissue ideally placed to resist vertical forces.

4 marks

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ii) Turgor pressure

Support as a result of flow of water.

- Parenchyma cells when filled with water offer support.
- Capillarity within the walls of xylem tissues results in to support
- Cohesion and adhesion forces within xylem tissue offer horizontal support to resist wind
- Transpiration pull offers a continuous stream of water from the roots to the xylem maintaining continuous support within the tissues.

Any 4,
4 marks

b) i) Muscular movement in fish;

- Myoline muscles are attached on either side of vertebral column preventing body shortening.
- Forward movement is by the lashing movement of the tail fin from side to side propelling the body forward.
- The streamlined shape of the fish help to reduce drag.
- Buoyancy is provided by the swim bladder in teleosts and the paired fins i.e. pectoral and caudal fin in elasmobranchs

Stability is provided by;

- vertical fins and lateral flattening of the body prevent yawing.
- Vertical and horizontal fins prevent rolling.
- Horizontal fins prevent pitching

Any 6
@ 1mark,
Total 6 marks

iii) Adaptations of fast moving animals

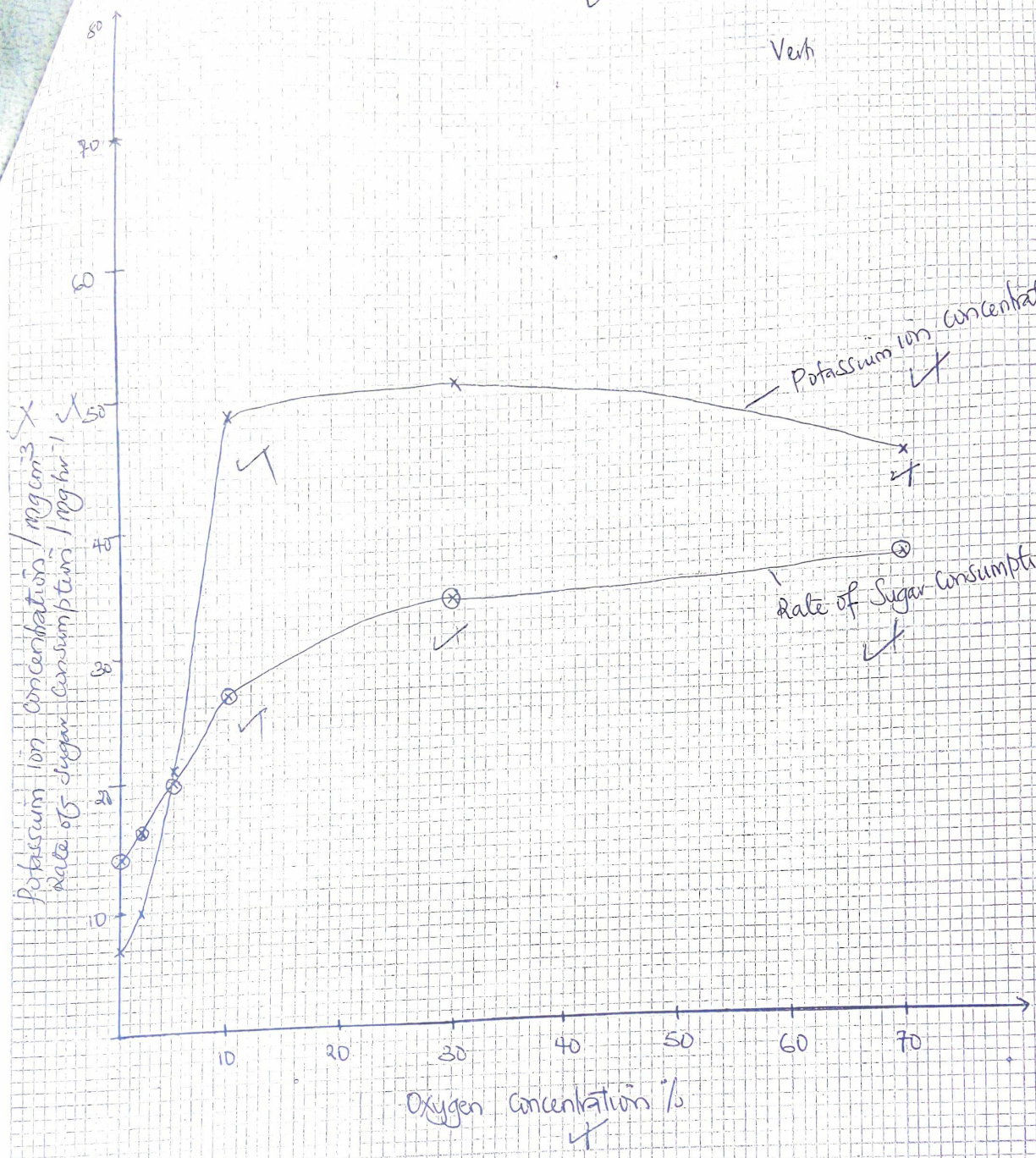
- high lever ratio of muscle to bone in legs i.e. long leg and short muscle attached to bone close to shoulder and hip joint.
- Many joints are moved in the same direction i.e. joint synergism.
- Elimination of other digits
- reduction of other digits
- confining muscle to where movement is least
- High concentration of muscles that are capable of sprinting but fatigue easily.

Any 6
@ 1mark
Total = 6 marks

END

~~ANSWER~~ ~~STH~~ GUIDE BIO P53072

Graph showing the effect of oxygen concentration on Potassium ion concentration and rate of sugar consumption in roots.



Vertical Scale

Potassium ion concentration - 1cm represent 5mg/cm³

Rate of Sugar Consumption 1cm represent 5mg/hr⁻¹

Horizontal Scale

1cm represent 5% oxygen concentration

T - 01
A - ~~02~~ 02½
S - 01½
P - 02
C - 01
K - 01

01